

STT 465: Bayesian Data Analysis

Homework 1: Probability Review

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Total: 5 points Due: September 25, 2026 at 5:00 PM

Instructions.

Show your work and justify your answers. Numerical answers without supporting calculations may receive little or no credit. You may use R for numerical calculations, but you should first write the relevant probability expression.

Question 1: Conditional Probability and a Partition [1 point]

A basketball team plays a game either at home or away. Let

$$H = \{\text{home game}\}, \quad W = \{\text{team wins}\}.$$

Suppose

$$P(H) = 0.60,$$

$$P(W \mid H) = 0.70,$$

and

$$P(W \mid H^c) = 0.45.$$

- (a) Use the definition of conditional probability to write an expression for $P(W \cap H)$ in terms of $P(W \mid H)$ and $P(H)$.

[0.2 pt]

- (b) Calculate

$$P(W \cap H)$$

and

$$P(W \cap H^c).$$

[0.2 pt]

- (c) Use the law of total probability to calculate

$$P(W).$$

[0.2 pt]

- (d) Calculate

$$P(H \mid W).$$

[0.2 pt]

- (e) Briefly explain why, in general,

$$P(H \mid W) \neq P(W \mid H).$$

[0.2 pt]

Question 2: Independence and Dependence

[1 point]

Let A and B be events satisfying

$$P(A) = 0.40, \quad P(B) = 0.50, \quad P(A \cap B) = 0.20.$$

- (a) Calculate

$$P(A \mid B).$$

[0.2 pt]

- (b) Calculate

$$P(B \mid A).$$

[0.2 pt]

- (c) Determine whether A and B are independent. Justify your answer in two different ways.

[0.3 pt]

- (d) Suppose now that

$$P(A \cap B) = 0.15.$$

Find the new value of

$$P(A \mid B)$$

and state whether A and B are independent.

[0.2 pt]

- (e) Explain why independence is a stronger condition than simply having a small value of $P(A \cap B)$.

[0.1 pt]

Question 3: Bayes' Theorem and Team Strength

[1 point]

Suppose a sports analyst classifies teams as either strong or weak.

Let

$$S = \{\text{team is strong}\},$$

and let

$$D = \{\text{team wins its first three games}\}.$$

Before observing the three games,

$$P(S) = 0.30, \quad P(S^c) = 0.70.$$

Suppose

$$P(D \mid S) = 0.80$$

and

$$P(D \mid S^c) = 0.20.$$

- (a) Calculate

$$P(D).$$

[0.2 pt]

- (b) Use Bayes' theorem to calculate

$$P(S \mid D).$$

[0.3 pt]

(c) Calculate

$$P(S^c \mid D).$$

Verify that your answers from parts (b) and (c) sum to one.

[0.2 pt]

(d) Suppose instead that the analyst begins with

$$P(S) = 0.60.$$

Without repeating all of the algebra, explain whether $P(S \mid D)$ will increase or decrease.

[0.2 pt]

(e) In one sentence, explain the roles of the *prior probability* and the *data* in this example.

[0.1 pt]

Question 4: Random Variables and the Binomial Distribution [1 point]

A basketball player makes a free throw with probability

$$p = 0.75,$$

and free-throw attempts are assumed to be independent.

Let

$$X_i = \begin{cases} 1, & \text{if the } i\text{th free throw is made,} \\ 0, & \text{otherwise.} \end{cases}$$

Let

$$Y = \sum_{i=1}^{10} X_i.$$

Thus,

$$Y \sim \text{Bin}(10, 0.75).$$

(a) What are

$$E[X_i] \quad \text{and} \quad \text{Var}(X_i)?$$

[0.2 pt]

(b) Use the binomial distribution to calculate

$$P(Y = 8).$$

[0.2 pt]

(c) Calculate

$$P(Y \geq 8).$$

You may use R for the numerical calculation.

[0.2 pt]

(d) Use linearity of expectation to derive

$$E[Y].$$

Do not simply quote the binomial formula.

[0.2 pt]

(e) Use independence to derive

$$\text{Var}(Y)$$

from the variances of the X_i 's.

[0.2 pt]

Question 5: Joint Distribution, Marginals, and Covariance [1 point]

Suppose a simplified basketball model records

X = number of three-point attempts

and

Y = number of three-point makes.

Their joint probability mass function is

	$Y = 0$	$Y = 1$
$X = 1$	0.30	0.20
$X = 2$	0.10	0.40

- (a) Verify that this table defines a valid joint probability mass function.

[0.2 pt]

- (b) Find the marginal pmf of X :

$$P(X = 1), \quad P(X = 2).$$

[0.2 pt]

- (c) Calculate

$$E[X] \quad \text{and} \quad E[Y].$$

[0.2 pt]

- (d) Calculate

$$E[XY].$$

[0.2 pt]

- (e) Use your answers to calculate

$$\text{Cov}(X, Y) = E[XY] - E[X] E[Y].$$

Are X and Y independent? Explain your answer.

[0.2 pt]